



The evolution of speech synthesis models

The TTS pipeline

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Introductions



Alessio Falai

- ❑ Currently working on speech generation models for Alexa+.
- ❑ In the past, implemented on-device TTS models for BMW/Mini and contributed to Amazon's speech-to-speech system Nova Sonic.
- ❑ Interested in cross-lingual synthesis and edge applications.



Alessandro Lombardi

- ❑ Currently working on automatic dubbing.
- ❑ In the past, focused on audio generation (music/sfx) models and speech synthesis for long-form narration.
- ❑ Interested in neural audio codecs and multi-modal applications.

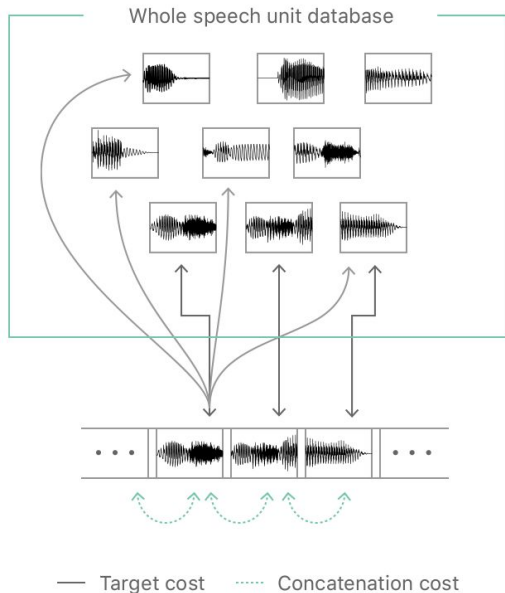


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(1) Modern and legacy TTS pipelines

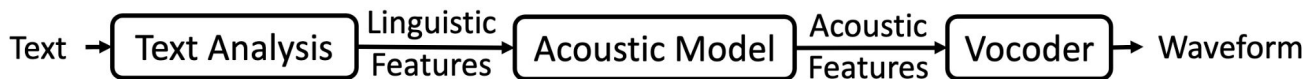
Legacy TTS systems



- ❑ Concatenative systems like **unit selection** are based on slicing recorded speech into “units” and then recombining them according to the input text.
 - ❑ The selection is based on two criteria: (1) the units must obey the target prosody and (2) the units should be concatenated without audible glitches at the unit boundary.
- ❑ Before concatenative systems, other solutions were explored, like **articulatory synthesis** which deals with the creation of physical models of human articulators.



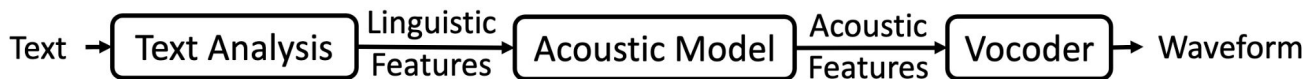
Modern TTS components



Raw text is taken as input and further processed by the **frontend** module.

- ❑ Text normalization and verbalization (*"this costs 10\$" → "this costs ten dollars"*)
- ❑ Homograph resolution (e.g. *"address"*)
- ❑ Prosody prediction (e.g. rising vs falling question intonation)
- ❑ Chunking (e.g. sentence splitting)
- ❑ Grapheme-to-phoneme conversion (*"chimneys" -> "tʃɪmɪnɪz"*).

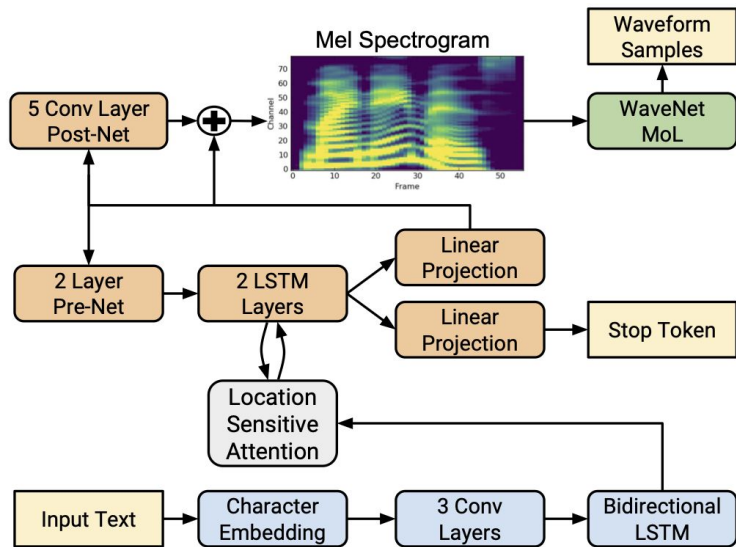
Modern TTS components



- ❑ Then, the **acoustic model** produces an intermediate speech representation.
 - ❑ The intermediate representation can be something like a spectrogram, discrete speech tokens or even unconstrained.
- ❑ This representation is finally converted to an audio waveform by the **vocoder**.

(2) Attention-based and non-autoregressive acoustic models

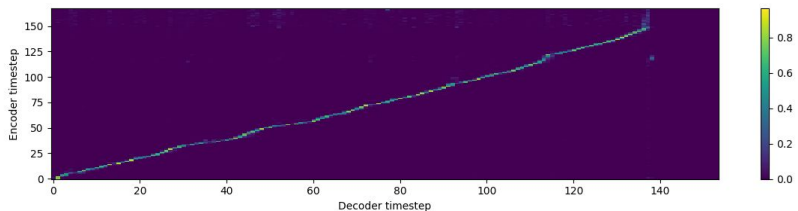
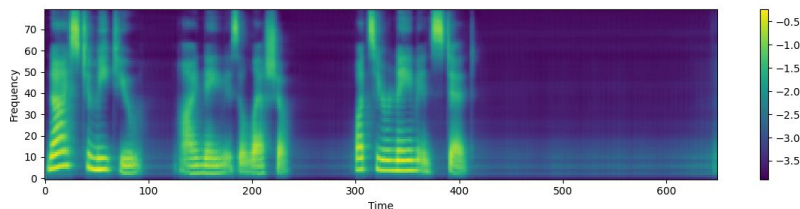
The Tacotron family of models



Tacotron is a **sequence-to-sequence** attention-based autoregressive model.

- ❑ The **encoder** (in blue) is a simple stack of convolutional layers followed by a bi-LSTM.
- ❑ The **decoder** (in orange) is formed of a recurrent network (LSTM-based), as well as:
 - ❑ **Pre-net**: an MLP acting as information bottleneck that is regarded as essential for learning attention.
 - ❑ **Post-net**: predicts a residual to add to the mel-spectrogram prediction to improve the overall reconstruction.

The attention mechanism in Tacotron



Plain sequence-to-sequence

- ❑ The encoder compresses the entire input sequence into a **context vector**.
- ❑ The decoder generates the output sequence relying on that same fixed context vector at every step.

Bahdanau attention (content-based attention)

- ❑ Each decoder step uses a new context vector built from a weighted mix of encoder states, which removes the single-vector information bottleneck.
- ❑ It also enables effective **alignment** between input and output elements (in speech it's learning character/phoneme **durations**).

Location-aware attention

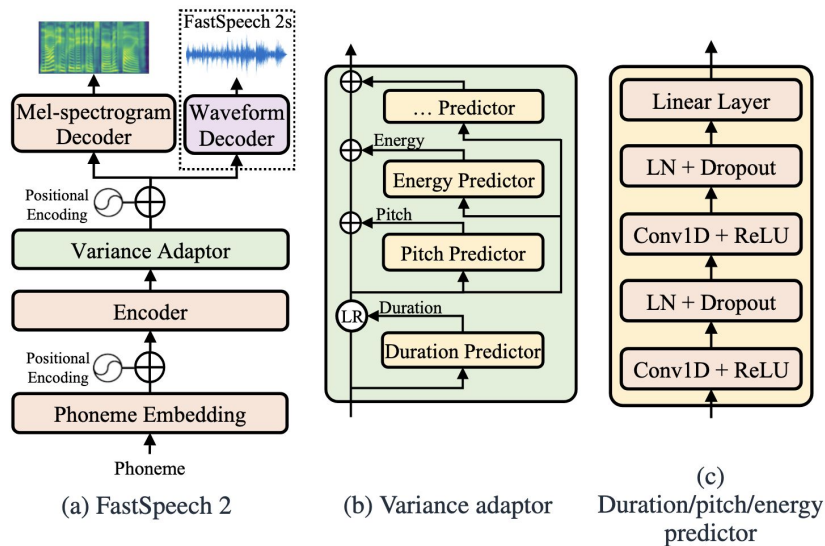
- ❑ Extends on Bahdanau attention to use **cumulative attention weights** from previous decoder steps as an additional feature.



Drawbacks of Tacotron attention

- ❑ Tacotron models suffer from **attention alignment problems** (like word skipping and repetition), especially at inference time.
 - ❑ This can be caused by out-of-the-domain words, long input texts, or intricacies of the target language.
- ❑ A lot of research has been focused on improving attention alignment.
 - ❑ Many solutions leverage the prior that text-speech alignment should follow a mostly **diagonal** (or, more generally, monotonic) path. That is, when one reads a text, it is natural to assume that the text position n progresses nearly linearly to the time t .
 - ❑ For example, in the “**guided attention loss**” paper the model gets penalized if the attention matrix is far from diagonal (e.g. reading the characters in random order).
 - ❑ Other work relies on **bidirectional decoding**, where an extra decoder takes the encoder outputs in the reverse order and a loss function compares the output states of the forward decoder with the backward one.

Alternatives to the attention module



Non-autoregressive decoder

- ❑ FastSpeech is a **Transformer**-based stack of multi-head self-attention (mirroring the encoder).
- ❑ Prevents **exposure bias** and error propagation issues from autoregressive models like Tacotron.
- ❑ Enables **parallel training and inference**, leading to major speed-ups over autoregressive models.

Explicit variance modelling

- ❑ Add variance information to the phoneme hidden sequence (to ease the **one-to-many mapping** problem in TTS).

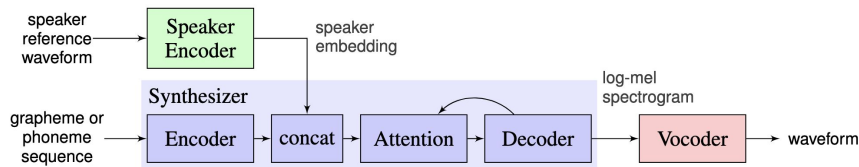


Duration modeling in FastSpeech

- ❑ The **duration predictor** takes the phoneme hidden sequence as input and predicts the duration of each phoneme, which represents how many mel frames correspond to this phoneme.
- ❑ The duration model is trained to predict ground truth durations extracted from either: (1) a pre-trained autoregressive model like Tacotron or (2) an external **forced aligner** like Montreal Forced Aligner (MFA).
 - ❑ Forced aligners like MFA rely on an ASR pipeline composed of specialised GMM/HMM models.

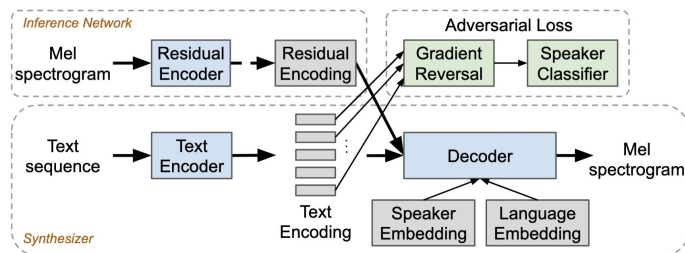
(3) Controlling speaker identity, language and style

Multi-speaker TTS



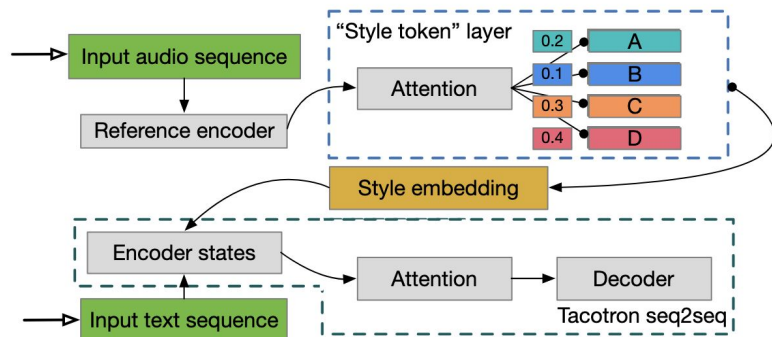
- ❑ A pre-trained **speaker encoder** extracts an embedding from a reference speech sample to condition the synthesis network.
- ❑ This embedding is concatenated with the encoder output at each time step to guide the generation towards the target speaker.
- ❑ An alternative approach uses a lookup table of fixed speaker embeddings, but this removes zero-shot synthesis.

Cross-lingual TTS



- Recordings from bilingual speakers are expensive to collect, i.e. each speaker in the training set speaks only one language, so **speaker identity is perfectly correlated with language**. This makes it difficult to transfer voices across different languages.
- The key factors for enabling **cross-lingual synthesis** (other than data and model scale) are: (1) input representations shared across languages, (2) careful training set construction and (3) auxiliary disentanglement guidance.

Controlling speaking styles



Training time

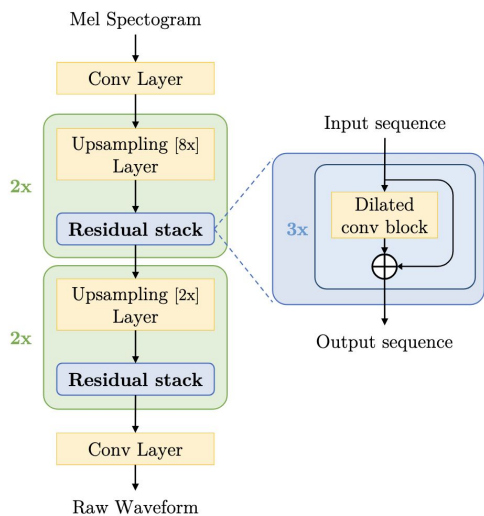
- ❑ The target spectrogram is fed to the reference encoder followed by a style token layer.
- ❑ The resulting **style embedding** is used to condition the Tacotron text encoder states.

Inference time

- ❑ We can feed an arbitrary reference signal to synthesize text with its speaking style.
- ❑ Alternatively, we can remove the reference encoder and directly control synthesis using the learned interpretable tokens.

(4) Neural vocoders and end-to-end-models

The generator in neural vocoders



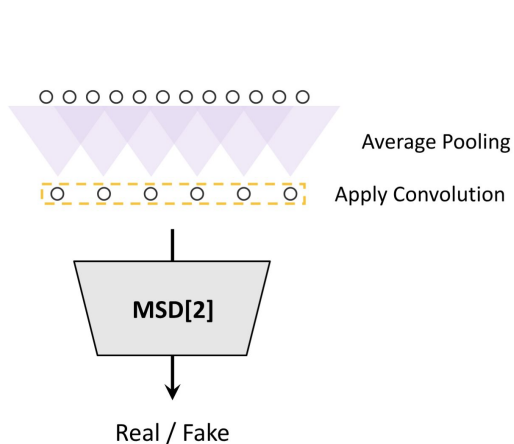
The adversarial game

- ❑ A **generator** attempts to produce samples that mimic the reference distribution (starting from noise).
- ❑ A **discriminator** tries to differentiate between real and generated samples.

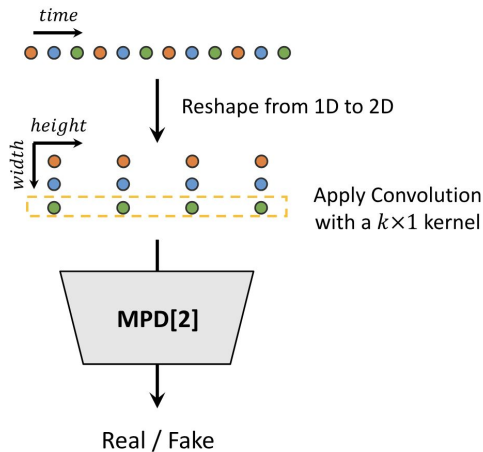
The **generator** is a fully convolutional feed-forward network with mel-spectrogram `s`` as input and waveform `w`` as output.

- ❑ Since `s`` is at lower temporal resolution than `w``, upsampling is performed via a stack of **transposed convolutions**.
- ❑ Each transposed convolutional layer is followed by a stack of **residual blocks** with dilated convolutions.

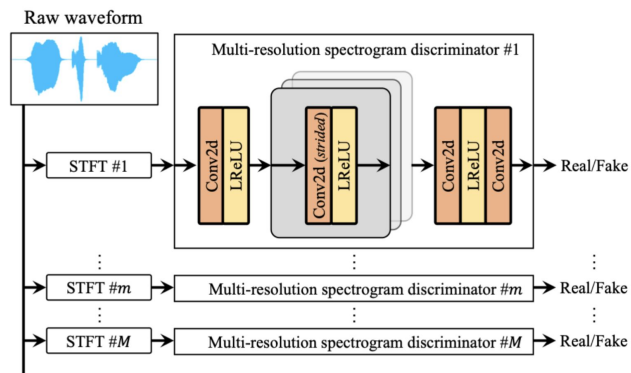
Discriminators in neural vocoders



Multi-scale

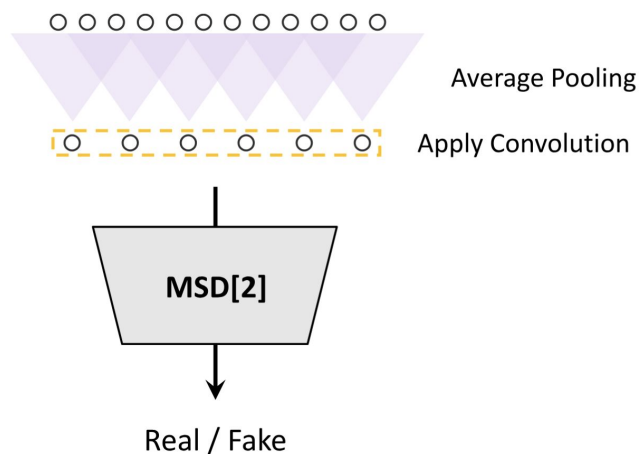


Multi-period



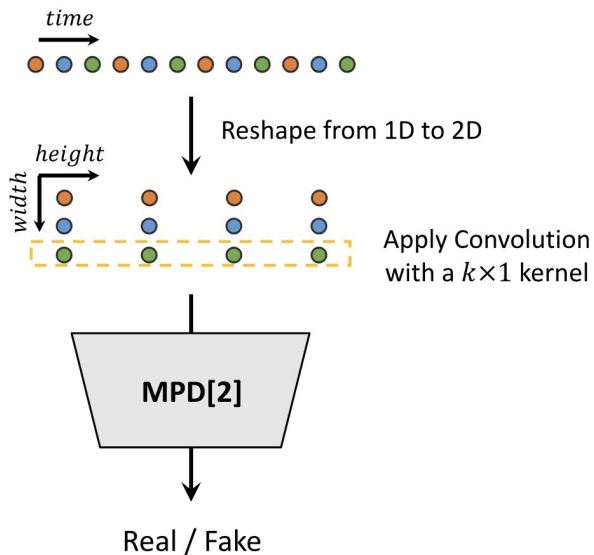
Multi-resolution

Multi-Scale Discriminators (MSD)



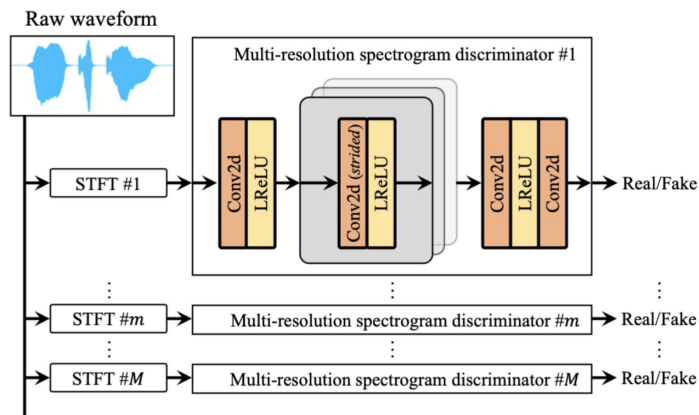
- ❑ Audio has structure at different levels: this structure has an inductive bias that each discriminator learns features for **different frequency range** of the audio.
- ❑ **3 discriminators** that have identical network structure but operate on different audio scales (1x/2x/4x).
- ❑ 2x/4x downsampled discriminators do not have access to high frequency component, so they are biased to learn discriminative features based on **low frequency components** only.

Multi-Period Discriminators (MPD)



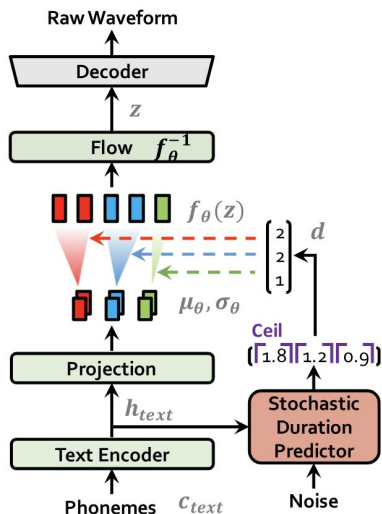
- As speech audio consists of **sinusoidal signals** with various periods, the diverse periodic patterns underlying in the audio data need to be identified.
- MPD is a mixture of sub-discriminators, each of which only accepts equally spaced samples of an input audio (a portion of **periodic signals**).

Multi-Resolution Discriminators (MRD)



- ❑ VAD research demonstrates that multiple features at different resolutions improve binary classification under various background noises.
- ❑ MRD borrows this idea and uses multiple linear spectrogram magnitudes computed using various STFT parameter sets to **operate on different spectral and temporal resolutions** simultaneously.

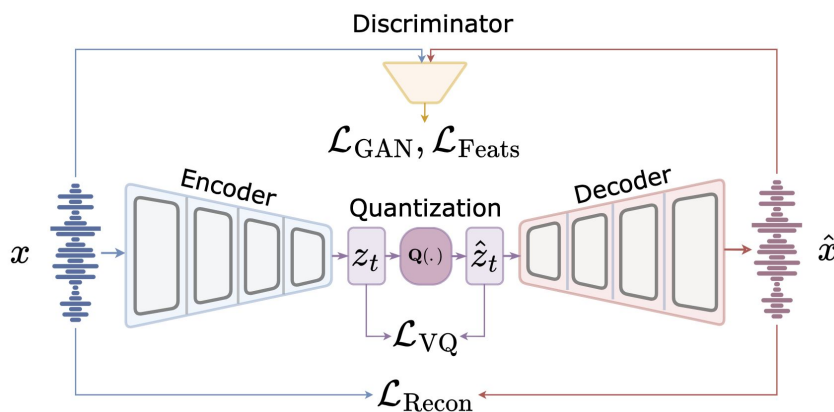
End-to-end models



- Traditional two-stage TTS (acoustic model + vocoder) suffers from **training-inference mismatch**, reducing audio quality.
- Some solutions include **fine-tuning vocoders** on synthetic outputs generated by TTS models.
- End-to-end models** like VITS learn to map directly from text to waveform, aiming to eliminate this mismatch by optimizing the whole pipeline jointly.

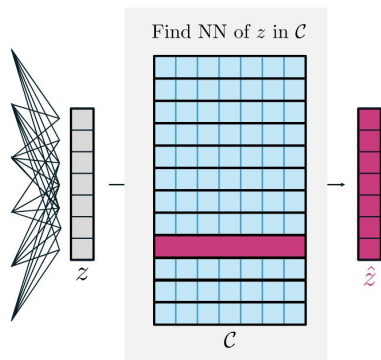
(5) Neural audio codecs

Overview of neural audio codecs

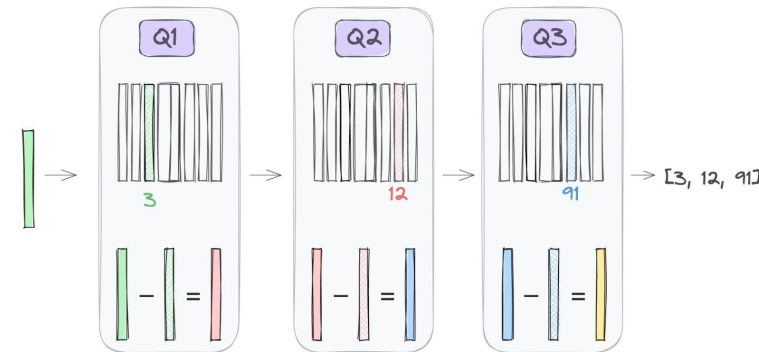


- ❑ The input signal is encoded into a latent representation, which is then discretized by a **quantizer**.
- ❑ The **decoder** reconstructs the signal from the quantized representations.
- ❑ Training typically involves a combination of reconstruction, adversarial, and vector quantization losses.
- ❑ The discrete tokens can then be used for **downstream modelling** tasks.

Quantization strategies

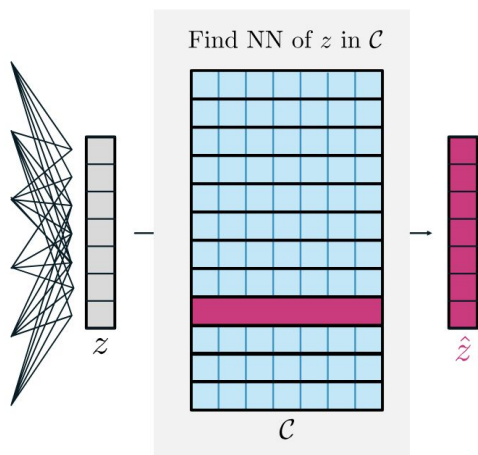


VQ



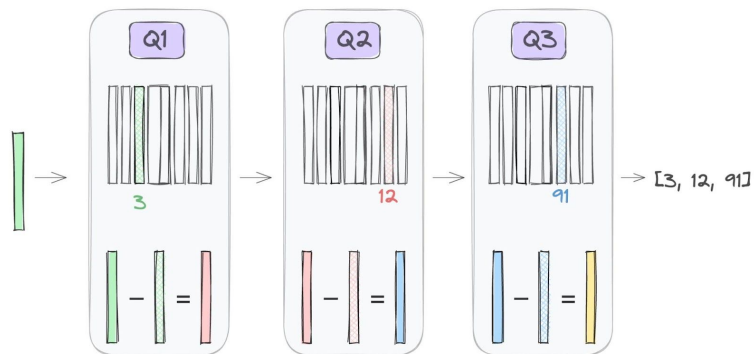
RVQ

Vector Quantization (VQ)



- ❑ The goal is to **compress** the encoder output by representing it with a small set of vectors (**codewords**).
 - ❑ The **codebook** is the set of codewords learned during training.
- ❑ The decoder learns to reconstruct using the **nearest neighbor** (L2 distance) codewords in the codebook.
- ❑ For downstream modeling, we take the **index (discrete)** of the nearest neighbor codeword, rather than the codeword itself.

Residual Vector Quantization (RVQ)



- ❑ Discretizes an embedding space by first performing VQ over a finite codebook.
- ❑ After that the **discretization errors** (the distance between the scalar vector and the nearest neighbor codeword embedding) are obtained and discretized again.
- ❑ This process continues for a predetermined **number of codebooks**.

Thank you for you attention
